The area of these triangles are

$\Delta ABC, \frac{1}{4}\ \Delta$ ABC, $\frac{1}{4}\times\frac{1}{4}\ \Delta\ ABC$,...

That is, $\Delta\ ABC, \frac{1}{4}\ \Delta\ ABC, \frac{1}{16}\ \Delta\ ABC$,...

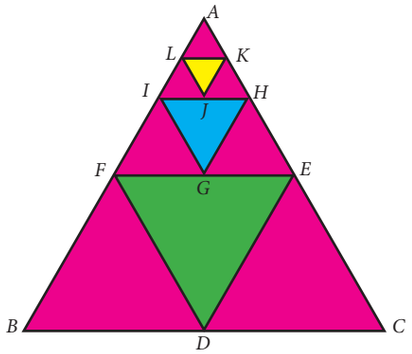
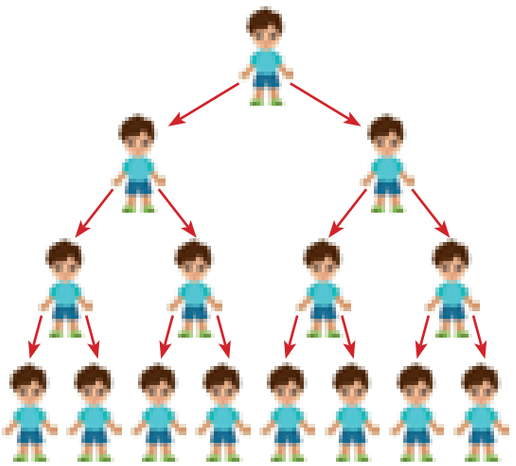


Fig.2.13

In this case, we see that beginning with ΔABC, we see that the successive triangles are formed whose areas are precisely one-fourth the area of the previous triangle. So, each term is obtained by multiplying $\frac{1}{4}$ to the previous term.


Fig.2.14

As another case, let us consider that a viral disease is spreading in a way such that at any stage two new persons get affected from an affected person. At first stage, one person is affected, at second stage two persons are affected and is spreading to four persons and so on. Then, number of persons affected at each stage are 1, 2, 4, 8, ... where except the first term, each term is precisely twice the previous term.

From the above examples, it is clear that each term is got by multiplying a fixed number to the preceding number.

This idea leads us to the concept of Geometric Progression.

Definition

A Geometric Progression is a sequence in which each term is obtained by multiplying a fixed non-zero number to the preceding term except the first term. The fixed number is called common ratio. The common ratio is usually denoted by r.

2.9.1 General form of Geometric Progression

Let a and $r \neq 0$ be real numbers. Then the numbers of the form $a, ar, ar^2, \ldots ar^{n-1}\ldots$ is called a Geometric Progression. The number 'a' is called the first term and number 'r' is called the common ratio.

We note that beginning with first term a, each term is obtained by multiplied with the common ratio 'r' to give $ar, ar^2, ar^3, \ldots$

2.9.2 General term of Geometric Progression

We try to find a formula for n^{th} term or general term of Geometric Progression (G.P.) whose terms are in the common ratio.

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